

Mohamed Matroud Abd El-Hady

Senior .NET Developer | Full-Stack Engineer

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PROFESSIONAL SUMMARY

Senior .NET Developer with 5+ years of professional experience designing, building, and scaling enterprise applications across ERP, HR, finance, and asset-management domains. Expert in ASP.NET Core, Web API, Entity Framework Core, and SQL Server, with full-stack capability using Angular and React. Proven track record delivering production systems for enterprise clients including Central National Bank and Jotun. Strong in modular architecture, RESTful API design, performance tuning, and translating complex business requirements into scalable, maintainable solutions.

TECHNICAL SKILLS

Languages & Runtime: C# 10/11, .NET 6 / 7 / 8 / 9 / 10, TypeScript, JavaScript (ES6+), T-SQL

Backend Frameworks: ASP.NET Core, ASP.NET MVC, Web API, Minimal APIs, Entity Framework Core, LINQ, Dapper

Architecture & Patterns: Clean Architecture, Modular Monolith, multi-tenant / split deployment, SOLID, DDD concepts, CQRS, Repository, Unit of Work, Mediator, Dependency Injection

API & Security: RESTful API design, JWT, OAuth 2.0, ASP.NET Identity, API versioning, Swagger / OpenAPI

Frontend: Angular, React 18, Vite, Material-UI, HTML5, CSS3, Bootstrap 5, jQuery, AJAX, JSON, responsive design

Databases: SQL Server (T-SQL, stored procedures, indexing, query optimization, schema design), EF Core migrations

Testing & Quality: xUnit, NUnit, Moq, unit & integration testing, code reviews

DevOps & Tooling: Git, GitHub, GitLab, Azure DevOps, Docker, IIS deployment, environment configuration management

Logging & Reporting: Serilog, custom audit logging, Crystal Reports, Stimulsoft Reports

Integration: Oracle integration, third-party API integration, real-time data synchronization, NetTopologySuite (spatial data)

PROFESSIONAL EXPERIENCE

Senior .NET Developer | Dynamic Business Solution

Mar 2025 – Present

- Lead backend architecture, API development, and database performance tuning for a multi-module enterprise ERP platform covering finance, inventory, procurement, and HR.
- Design and implement new ERP modules using modular architecture principles, applying SOLID and proven design patterns to ensure scalability and maintainability.
- Modernize and harden legacy APIs to improve performance, security, and reliability through targeted refactoring and query optimization.
- Build RESTful APIs with ASP.NET Core and Web API using dependency injection and clean separation of concerns.
- Design and optimize SQL Server databases including stored procedures, indexes, and schema tuning for high-throughput operations.
- Contribute to architectural planning, requirements analysis, and technical decisions for reusable, extensible modules.

Project: Nova HR System

Scope: Full-featured HR platform managing attendance, leave, payroll, onboarding, and transfers.

Stack: ASP.NET Core, Web API, SQL Server, Entity Framework Core

Contribution: Enhanced features, resolved critical defects, modeled core domain entities, and authored business logic.

Project: Enterprise ERP Platform (Built from Scratch)

Scope: Complete ERP solution spanning finance, inventory, procurement, and HR business functions.

Stack: ASP.NET Core, Web API, SQL Server, Entity Framework Core, Git

Contribution: Defined data models, designed backend logic, built APIs, and supported frontend integration.

Project: Nova Visitors — Multi-Tenant Community Access & Owner Management Platform

Scope: Full-stack platform for gated communities managing owner registration, resident authentication, unit / person / visit records, and administrative dashboards. Split deployment — a per-village server holding local data plus a shared central registry syncing owner identities and cross-village reporting over HTTP, with each village isolated to its own database.

Stack: .NET 8, ASP.NET Core Web API, EF Core 8, SQL Server 2022, ASP.NET Identity + JWT, React 18, Vite, Material-UI, NetTopologySuite, Docker

Contribution: Backend and frontend developer — designed the split-deployment dependency injection, built the /CentralOwnerSync HTTP surface, added the persons filter page, and delivered the bilingual RTL rework.

Full-Stack Developer | Ultimate Solutions

Nov 2023 – Mar 2025

- Delivered end-to-end software solutions spanning frontend and backend for finance and restaurant-management clients, including systems deployed in the Saudi Arabia market.
- Designed and developed RESTful services using ASP.NET Core and Web API, applying OOP principles and design patterns to maintain clean, testable code.
- Performed relational database design and management using SQL Server, including stored procedures and query optimization.
- Integrated Angular, jQuery, AJAX, and JSON with backend services to deliver a seamless user experience.
- Used Entity Framework and LINQ for data access, manipulation, and complex query design.
- Contributed to the design of cross-system APIs enabling reliable integration across multiple platforms.

Project: E-Invoice Integration Platform — Saudi Arabia

Scope: Customer invoice creation and management platform integrated with a national real-time tracking system for tax compliance.

Stack: ASP.NET Core, Web API, SQL Server, Angular, Bootstrap, Stimulsoft Reports

Contribution: Owned frontend development and API design, ensuring reliable data flow between the platform and external regulatory systems.

Project: Restaurant Management System Integration

Scope: Integration of a restaurant POS with centralized accounting and inventory platforms.

Stack: ASP.NET Core, Web API, SQL Server

Contribution: Designed and implemented APIs enabling real-time synchronization of sales, accounting, and inventory data — improving operational visibility.

.NET Developer | GBS — Global Barcode Solution

Feb 2021 – Oct 2023

- Delivered end-to-end enterprise systems for asset management, warehouse control, and operations across web, mobile, and desktop platforms for clients including Central National Bank and Jotun.
- Developed and maintained full-stack solutions using ASP.NET, ASP.NET Core, MVC, Telerik UI, SQL Server, Web API, AJAX, and Crystal Reports.
- Designed RESTful APIs supporting cross-platform integration between web, desktop, and mobile clients.
- Conducted database design and optimization for enterprise-grade SQL Server deployments.
- Integrated external systems including Oracle into custom .NET applications.
- Collaborated with cross-functional teams to translate business requirements into effective technical solutions, troubleshooting full-stack issues to ensure reliability.

Project: Asset Auditing System — Central National Bank

Scope: Enterprise asset-management solution tracking organizational assets across web, mobile, and desktop applications.

Stack: ASP.NET Core, MVC, Telerik UI, Web API, SQL Server, Crystal Reports

Contribution: Full-stack development, API design, and cross-system integration across three interconnected client platforms.

Project: Warehouse Control System — Jotun

Scope: QR / barcode design and warehouse operations management across web and desktop.

Stack: ASP.NET Core, MVC, Telerik UI, Web API, SQL Server, Crystal Reports

Contribution: Owned full-stack development across the web and desktop applications.

EDUCATION

Bachelor of Computer Science | Thebes Academy, Egypt | Graduated 2019

Graduation Project: Pharmacy Management System — rated Excellent.

CERTIFICATIONS & TRAINING

- Web Development Diploma (C#, ASP.NET) — Russian Cultural Center.
- Frontend Design Diploma — National Telecommunication Institute (NTI).
- International English Language Certificate — Ain Shams University (M.E.S.C / Arab University Academy), 2018.

LANGUAGES

- Arabic — Native
- English — Professional working proficiency

SOFT SKILLS

Problem-solving and critical thinking • Strong written and verbal communication • Adaptable and team-oriented • Performs well under deadlines and pressure • Collaborative cross-functional contributor